Hybrid Deterministic + LLM Extraction of Structured Seizure Phenotypes

An extended narrative literature review centred on
ExECT annotations and synthetic seizure-frequency benchmarks

Core thesis

The strongest publishable framing is not simply that a large language model can read epilepsy clinic letters. It is that a clinically grounded, evidence-preserving, schema-constrained hybrid pipeline can extract seizure phenotypes from narrative letters more transparently than an LLM-only system and more flexibly than a deterministic-only system. The benchmark pair you have chosen - the public ExECT annotation corpus and the synthetic seizure-frequency corpus - is unusually powerful because it forces the dissertation to confront two different forms of structure: broad clinical concept annotation and narrow but difficult temporal-normalised seizure-frequency labelling.

Prepared for MSc AI dissertation planning. This is a narrative review for study and design thinking, not a systematic review. Full clickable source links are included in the bibliography.

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Contents

1 How to read this review

This review is written as a working companion for a dissertation rather than as the compact related-work section of a conference paper. Its purpose is to help you see the argumentative landscape: what each paper is really claiming, what assumptions sit underneath the methods, which methodological decisions are likely to impress or irritate reviewers, and where your own hybrid deterministic plus LLM pipeline can make a credible contribution.

The centre of gravity is your chosen benchmark pair. Fonferko-Shadrach, Lacey, Akbari, et al. (2024) provide a public, clinically authored synthetic letter corpus with detailed annotation guidelines and an ExECTv2 benchmark. Gan, Roy, et al. (2026) propose a more recent synthetic-data route for seizure-frequency information extraction, pairing synthetic NHS-style letters with structured labels, rationales, and evidence spans, then testing whether models trained only on synthetic data generalise to clinician-checked real letters. These papers are not competitors in a simple sense. They represent two different answers to the same underlying problem: how can epilepsy NLP research become reproducible, clinically meaningful, and shareable when the data are narrative, private, heterogeneous, and full of temporal ambiguity?

The review therefore moves through three layers. First, it examines the clinical and informatics problem: why seizure phenotypes are trapped in clinic letters, and why seizure frequency is especially difficult. Second, it reads the two benchmark papers closely in context, treating them as methodological resources rather than just citations. Third, it synthesises the literature into a design language for your dissertation: what should be deterministic, what should be delegated to an LLM, how to evaluate the hybrid pipeline, and which ambiguities should be made explicit rather than hidden.

A useful reading stance

Do not ask only, "Which model has the highest F1?" Ask: "What kind of clinical truth is the system trying to extract? What evidence is the output grounded in? What does the schema make visible, and what does it suppress? What kinds of mistakes would be clinically tolerable, and what kinds would be unacceptable?"

2 The clinical object: seizure phenotypes as documented facts

Epilepsy clinic letters are not merely administrative correspondence. They are compact clinical narratives that often contain the most important longitudinal facts about a patient's epilepsy: diagnosis, epilepsy type, seizure type, semiology, frequency, date of last seizure, medication history, investigations, side effects, comorbidities, and plans. Much of this material is not reliably stored in structured fields, and even when structured fields exist they may lag behind the nuanced account contained in free text. This is the central motivation behind ExECT and the broader epilepsy NLP literature (Fonferko-Shadrach, Lacey, White, et al. 2019; Yew et al. 2023; Xie et al. 2023b).

The clinical difficulty is that a seizure phenotype is not a single label. A useful phenotype is a temporally anchored, assertion-qualified, clinically interpreted bundle of information. A letter may say that the patient had focal impaired awareness seizures in childhood, now has rare nocturnal convulsions, has weekly auras, is seizure-free from disabling seizures, and remains on lamotrigine while levetiracetam was stopped because of mood effects. A flat extractor that returns only "focal seizure" or "monthly" would miss much of the clinical meaning.

For a dissertation framed around hybrid deterministic and LLM extraction, this matters because it changes the extraction target. You are not extracting the metaphysical truth of the patient's epilepsy. You are extracting the phenotype as documented in the letter, ideally with evidence spans, temporality, assertion status, and uncertainty. That distinction is not pedantic. It is what separates a responsible clinical NLP system from an overconfident summariser.

2.1 ILAE terminology: necessary but not sufficient

The International League Against Epilepsy provides the terminological scaffold for seizure and epilepsy classification. The 2017 seizure-type classification distinguishes focal, generalised, and unknown onset categories and encourages more descriptive seizure naming (Fisher et al. 2017). The 2017 classification of the epilepsies situates seizure type within a broader hierarchy that includes epilepsy type, epilepsy syndrome, and aetiology (Scheffer et al. 2017). The 2022 ILAE glossary further expands the language of seizure semiology (Blumcke et al. 2022). The 2025 updated classification continues this process by sharpening the distinction between classifiers and descriptors and by revising aspects of terminology such as consciousness descriptors (Beniczky, Wirrell, et al. 2025).

These standards are essential, but a clinic-letter extractor cannot simply enforce textbook language. Real letters contain legacy terms such as "complex partial seizure," lay terms such as "blackouts," ambiguous words such as "episodes," copied-forward diagnoses, abbreviations, and mixed clinician habits. A good extraction schema therefore needs two layers: a controlled output vocabulary and a record of the original surface text. The mapping is itself part of the scientific contribution.

A hybrid pipeline can exploit this distinction. Deterministic components can maintain dictionaries, synonym maps, ILAE version mappings, medication lexicons, section boundaries, and temporal rules. The LLM can interpret messy language, long-distance relations, and semiological paraphrases, but should be required to output controlled labels and evidence. This is where the dissertation framing becomes sharper than generic LLM extraction.

Schema principle

Every clinically important extraction should have at least five attributes: a controlled label, the original wording or evidence span, assertion status, temporality, and uncertainty. For frequency and dates, add a normalised value and an anchor date. Without these attributes, performance metrics can look good while the output remains clinically brittle.

3 The ExECT lineage: from practical rule-based extraction to public annotation

The ExECT work is the most important historical anchor for your dissertation because it demonstrates that epilepsy-specific clinic-letter extraction has already been pursued in a serious clinical NLP tradition. The 2019 ExECT paper developed a system in GATE that combined rule-based and statistical methods to extract structured epilepsy data from clinic letters, including diagnosis, epilepsy type, seizure type, seizure frequency, anti-seizure medication, and investigations (Fonferko-Shadrach, Lacey, White, et al. 2019). Its reported performance was clinically promising but uneven across fields: medication extraction was strong, while seizure frequency and some seizure-type categories had lower recall.

That unevenness is not a flaw to dismiss. It is a clue. Structured medication statements are often written in repeatable forms: drug name, dose, unit, frequency, action. Seizure frequency is more narrative: "about once every few months," "two clusters since last review," "no definite seizures but occasional funny feelings," "one nocturnal episode after missing tablets." The ExECT lineage already shows that the apparent simplicity of the phenotype field is misleading. Some parts of the letter behave like forms; others behave like stories.

3.1 The 2024 annotation paper: what it contributes beyond data

Fonferko-Shadrach, Lacey, Akbari, et al. (2024) is easy to underestimate if it is read merely as "the paper that releases synthetic annotated letters." Its deeper contribution is epistemic. It asks what it means to create a clinically credible gold standard for epilepsy NLP when real letters cannot be openly shared and when even trained annotators disagree.

The paper describes 200 synthetic UK outpatient epilepsy clinic letters written by neurology consultants, specialist trainees, and epilepsy specialist nurses. The letters are synthetic rather than de-identified real letters; they were based on the style and content of outpatient consultations but contain no real patient information. The dataset was double annotated using Markup and an epilepsy concept list based on UMLS, with subsequent review and consensus to produce a gold standard (Fonferko-Shadrach, Lacey, Akbari, et al. 2024; Fonferko-Shadrach, Lacey, and Pickrell 2023).

The annotation schema is broad. It covers birth history, epilepsy diagnosis, epilepsy type or syndrome, seizure type, epilepsy cause, investigations such as EEG, CT and MRI, onset, patient history, current anti-seizure medication with dose information, seizure frequency, and date or timing of diagnosis. It also includes certainty attributes for selected diagnosis and history concepts. This breadth matters because it frames epilepsy phenotyping as more than frequency extraction. It is closer to a clinic-letter phenotype substrate.

The reported inter-annotator agreement is revealing. Overall agreement was moderate rather than perfect, with some entities much easier than others. Prescriptions were comparatively reliable; "when diagnosed" and other temporally or narratively embedded concepts were harder. ExECTv2 achieved high overall scores against the consensus standard, but seizure frequency again remained one of the weaker categories (Fonferko-Shadrach, Lacey, Akbari, et al. 2024). The message is not simply that NLP is hard. It is that the gold standard is itself a clinical-linguistic artefact, shaped by guideline clarity, concept boundaries, and adjudication decisions.

What the ExECT annotation paper should make you ask

When your system disagrees with the benchmark, is the system wrong, the annotation under-specified, or the clinical sentence genuinely ambiguous? A sophisticated dissertation should not treat the gold standard as infallible; it should treat it as the best available operationalisation of a difficult clinical reading task.

3.2 Synthetic letters in ExECT: privacy solution or domain approximation?

The ExECT annotation corpus solves a practical problem: real clinical letters are difficult to share. The public dataset gives researchers something to benchmark against, inspect, and reproduce. But the synthetic nature of the letters introduces a second-order question: do systems that perform well on clinician-authored synthetic letters perform equally well on real letters? This is not a reason to avoid the corpus. It is a reason to be precise about what the corpus tests.

The ExECT synthetic letters appear especially valuable for testing broad extraction schema implementation, annotation-guideline fidelity, concept coverage, and reproducible evaluation against a public gold standard. They are less obviously sufficient for testing robustness to genuine institutional drift, messy copy-forward artefacts, unusual clinician phrasing, dictation errors, rare syndromes, and complex longitudinal contradictions. In other words, they are excellent for benchmark transparency but not a substitute for external validation.

This distinction is directly relevant to your hybrid design. A deterministic pipeline may look strong on ExECT because the schema, vocabulary, and annotation conventions are well aligned. An LLM may look strong because the letters are coherent and clinically written. The dissertation should therefore use ExECT not as the final proof of real-world deployment, but as a public benchmark for reproducible method comparison.

3.3 Why ExECT is ideal for a hybrid dissertation

ExECT gives you a broad, clinically meaningful extraction playground. It includes fields that deterministic methods should handle well, such as medication names and investigation mentions, and fields where LLM interpretation may add value, such as seizure description, history, uncertain diagnosis, and frequency. This is exactly what a hybrid architecture needs: heterogeneous subproblems that should not all be solved with the same tool.

A deterministic-only baseline can be credible because ExECT itself is rooted in deterministic and rule-based clinical NLP. An LLM-only baseline can test whether modern models can subsume some of that engineering. The hybrid system can then argue for complementarity: deterministic components retrieve candidate evidence, normalise outputs, enforce schema validity, and flag contradictions, while the LLM resolves semantic ambiguity and generates structured interpretations from messy text.

A productive ExECT experiment

Do not evaluate only "overall extraction F1." Break the ExECT labels into families: structured medication/investigation fields, diagnostic labels, seizure-type labels, temporality-sensitive fields, and frequency fields. Then ask where the hybrid system actually improves the error profile. A small improvement in the difficult family may be more publishable than a large improvement in an easy field.

4 Seizure frequency as the stress test

Seizure frequency looks like a numeric extraction problem until one reads real letters. Then it becomes obvious that it is a temporal reasoning problem, an event-typing problem, and sometimes a clinical uncertainty problem. The literature repeatedly returns to seizure frequency because it is both clinically central and linguistically awkward.

Frequency can be explicit: "two seizures per month." It can be approximate: "about once every few weeks." It can be interval-based: "three since January." It can be negative: "no seizures since surgery." It can be event-type specific: "auras weekly, convulsions once a year." It can be embedded in clusters: "two clusters of three to four seizures." It can be unknown: "not clear how often these occur." It can also be absent, which is not the same as unknown. These distinctions are central in Gan, Roy, et al. (2026), but they are foreshadowed throughout the earlier literature.

Xie et al. (2022) approached seizure outcome extraction as a machine-reading task: given a clinical note, answer questions about whether the patient had recent seizures, how often they occurred, and when the most recent seizure happened. This framing was important because it shifted the task from simple mention detection to answer extraction. The models needed to identify not just a seizure term, but the clinically relevant answer span.

Decker et al. (2022) developed a rule-based algorithm for extracting seizure types and frequencies. The paper is particularly valuable because it highlights generalisation failure: acceptable internal performance did not carry cleanly to an external centre. That finding should be on the first page of your mental methods checklist. Any dissertation that claims clinical utility from a single-source evaluation needs to temper its claims. Local note style is not a nuisance variable; it is part of the problem.

Fernandes, Goldenholz, et al. (2024) extended the seizure-control extraction line by combining a pretrained RoBERTa question-answering model with regular expressions to convert extracted text into structured seizure-control categories. This is already a hybrid idea, even if it does not use a general-purpose LLM as the central semantic component. The paper shows that post-processing model outputs into clinically meaningful categories is not an afterthought. It is a core part of making extracted text usable.

The LLM-era papers sharpen the same message. Holgate et al. (2025) focus specifically on fine-tuning LLMs for seizure-frequency extraction and normalising frequency into seizures per month. Abeysinghe et al. (2025) evaluate pretrained language models and generative LLMs for structured seizure-frequency extraction from epilepsy evaluation reports, showing that direct extraction from long text can be vulnerable to false positives and that task decomposition matters. Ojemann, Xie, et al. (2025) show that zero-shot prompting can be useful, but also that fine-tuned task-specific models remain strong comparators. Fang et al. (2025) bring LLM extraction into epilepsy clinic letters more directly, reporting encouraging performance for epilepsy type, seizure type, current medication, and symptoms, but also reinforcing the need for careful task design.

The frequency trap

A system that converts "seizure-free since 2022 except rare auras" into a single monthly seizure rate has not solved the clinical problem. It has collapsed it. Ask whether frequency is being extracted per seizure type, per disabling event, per all events, or per the clinician's implicit outcome category.

5 The Gan synthetic-letter paper: why it is more than data augmentation

Gan, Roy, et al. (2026) is best understood as a response to three intertwined problems: privacy, reproducibility, and label scarcity. Real seizure-frequency data are clinically valuable but difficult to share. Manual annotation is costly because frequency expressions require temporal interpretation. If the field relies only on private local datasets, results remain hard to replicate. Gan and colleagues propose a route around this bottleneck: generate fully synthetic, task-faithful, NHS-style clinical letters paired with normalised seizure-frequency labels, rationales, and evidence spans, then use these synthetic letters to train open-weight models and evaluate on held-out clinician-checked real letters.

The paper's core claim is not merely that synthetic text can improve model performance. It is that synthetic supervision can be made sufficiently faithful to a specific clinical extraction task to support reproducible model development without distributing patient text. That is a stronger and more interesting claim. It depends on the label scheme, the generation process, the evidence spans, the rationale design, and the real-letter evaluation.

5.1 The label scheme as argument

Gan et al.'s seizure-frequency categories are central to the paper. The fine-grained scheme expresses rates in seizures per month using clinically meaningful boundaries, while the pragmatic scheme collapses those labels into broader categories. The inclusion of explicit "unknown" and "no seizure" categories is methodologically important. It acknowledges that an extraction system must distinguish between no documented seizures, unclear frequency, and no relevant information. Many poor extraction systems blur these cases.

This label design should influence your own schema. A clinical phenotype extractor should not treat "unknown" as a failure state to be hidden. It can be a valid output, especially when the letter does not support a more specific answer. Similarly, "no seizure" should be grounded in explicit evidence rather than inferred from silence. A deterministic validator can help enforce this distinction by requiring evidence for absence claims and by flagging unsupported frequency values.

5.2 Rationales and evidence spans

One of the most dissertation-relevant aspects of Gan, Roy, et al. (2026) is the use of rationale and evidence-aware supervision. The models are not trained only to emit a final class; they are trained with intermediate explanations or evidence-oriented outputs that identify seizure-related spans, interpret temporal information, and produce the normalised label. Even if the final evaluation focuses on labels, the design recognises that clinicians need to verify why the system produced an output.

This aligns closely with your hybrid framing. Evidence spans are the bridge between deterministic retrieval and LLM semantic interpretation. The deterministic system can find candidate frequency-bearing sentences, seizure terms, dates, and negation cues. The LLM can interpret them in context. The final validator can check that the output is supported by the cited evidence. This turns the LLM from an oracle into a constrained reader.

5.3 Synthetic supervision and open-weight deployment

Gan et al. emphasise open-weight models that can be run inside hospital systems. This matters for clinical NLP because data governance often makes cloud-based processing difficult or impossible. A dissertation that compares local open models with API-based LLMs could frame the comparison not only around accuracy but around deployability, privacy, reproducibility, and cost.

The paper reports that models trained entirely on synthetic data generalised to real clinician-checked letters with encouraging performance, particularly when trained with larger synthetic datasets and structured outputs (Gan, Roy, et al. 2026). But the paper also has limitations: it is a preprint, it comes from a single UK centre context, the synthetic corpus may reflect template constraints, and prospective clinician user studies remain future work. These limitations are not weaknesses to exploit; they are openings for your dissertation to be precise about scope.

What Gan gives your dissertation

Gan gives you a benchmark for the hardest phenotype subtask: seizure frequency. It also gives you a vocabulary for evidence-grounded supervision, synthetic data realism, label design, and open-weight deployment. The opportunity is to show whether a hybrid deterministic plus LLM pipeline can use those ideas without becoming a black-box synthetic-data exercise.

6 The benchmark pair: why ExECT plus Gan is unusually strong

The two benchmark papers complement each other almost perfectly for your framing. ExECT is broad, public, clinically authored, and annotation-guideline driven. Gan is narrow, synthetic-supervision driven, evidence-aware, and focused on the difficult normalisation problem of seizure frequency. One tests breadth of epilepsy phenotyping; the other tests depth of a high-value temporal phenotype.

Table 1: How the two benchmark papers support different parts of a hybrid dissertation.

Dimension	Fonferko-Shadrach et al. 2024	Gan et al. 2026
Data object	Publicly available synthetic epilepsy clinic letters authored by clinical professionals and annotated with a broad epilepsy concept list (Fonferko-Shadrach, Lacey, Akbari, et al. 2024; Fonferko-Shadrach, Lacey, and Pickrell 2023).	Fully synthetic NHS-style letters generated for seizure-frequency extraction, paired with structured labels, rationales, and evidence spans; evaluated against clinician-checked real letters (Gan, Roy, et al. 2026).
Primary task	Broad clinical concept extraction: diagnosis, seizure type, frequency, medications, investigations, history, onset, cause, and related attributes.	Fine-grained and pragmatic seizure-frequency classification/normalisation, including explicit rates, ranges, seizure-free statements, unknowns, and no-seizure categories.
Main methodological value	Shows how to operationalise an epilepsy letter annotation schema and how difficult clinical annotation can be even with guidelines.	Shows how synthetic supervision can encode a difficult temporal reasoning task and support reproducible model development without sharing patient text.

Dimension	Fonferko-Shadrach et al. 2024	Gan et al. 2026
Best use in your dissertation	Benchmark broad extraction, schema mapping, deterministic/LLM/hybrid comparison, and annotation compatibility.	Benchmark the hardest temporal-normalisation component and test evidence-grounded output design.
Main caveat	Synthetic letters may not fully reflect messy real-world distribution shift. The benchmark tests reproducibility and schema fidelity more than external deployment.	Preprint status and single-context real evaluation mean claims should be treated carefully. Synthetic generation may shape the error distribution.

The pair also gives you a clean two-level evaluation story. Level one: can the system extract a broad set of epilepsy phenotype variables from public annotated letters? Level two: can it handle seizure frequency with enough precision, evidence grounding, and uncertainty handling to be clinically meaningful? This is more compelling than building a pipeline and testing it on one monolithic label set.

7 From methods literature to design logic

The wider literature suggests a consistent pattern: neither deterministic rules nor LLMs alone are sufficient across all components of epilepsy phenotype extraction. Deterministic methods are precise, auditable, and excellent for lexical regularities, but brittle under paraphrase and institutional variation. Transformer and LLM methods are more flexible, but they can over-infer, hallucinate unsupported answers, mishandle temporality, and obscure the reason for an output. The hybrid design is therefore not a compromise; it is a division of labour.

Table 2: Methodological lessons from the literature for a hybrid pipeline.

Approach	What it does well	Where it struggles	Representative sources
Rule-based clinical NLP	High precision for known patterns, medication formats, section headings, negation phrases, and controlled vocabularies. Easy to inspect and debug.	Poor robustness to paraphrase, new templates, implicit relations, and external institution style.	Fonferko-Shadrach, Lacey, White, et al. (2019); Decker et al. (2022).
Transformer machine reading	Strong span extraction and task-specific answer finding when annotated examples exist. Better than simple dictionary matching for frequency/date questions.	Requires labelled data; may not generalise cleanly across contexts; answer spans still need normalisation.	Xie et al. (2022); Xie et al. (2023a).

Approach	What it does well	Where it struggles	Representative sources
LLM prompting	Flexible semantic interpretation, few-shot adaptation, and broader context understanding. Useful where annotation is sparse.	Can over-read, produce unsupported outputs, vary with prompt, and struggle with long noisy text.	Fang et al. (2025); Ojemann, Xie, et al. (2025); Abeysinghe et al. (2025).
Synthetic-data fine-tuning	Supports reproducible training without distributing real patient text; can target rare linguistic cases by design.	Synthetic distributions may encode generator biases and omit real-world messiness. Requires real evaluation.	Gan, Roy, et al. (2026); J. J. Li et al. (2021); Melamud and Shivade (2019).
Hybrid deterministic + LLM extraction	Combines evidence retrieval, schema validation, semantic interpretation, and auditability. Can reduce unsupported outputs.	More engineering complexity; ablation is needed to prove that each component adds value.	Fernandes, Goldenholz, et al. (2024); Liu et al. (2025); Dao et al. (2025).

The hybrid argument is strongest when it is operational rather than rhetorical. You should be able to point to a specific error and say which component should prevent it. For example, if an LLM extracts "monthly seizures" from a sentence about childhood seizures, the deterministic temporal filter or assertion validator should flag the evidence as historical. If a rule system misses "episodes of blankness and lip-smacking," the LLM should map it to a candidate seizure description while preserving the original wording and uncertainty. If a frequency answer has no evidence span, the validator should reject it or mark it for human review.

8 What should be deterministic? What should the LLM do?

A useful hybrid architecture starts by refusing to use the LLM for tasks that do not need it. This reduces cost, improves reproducibility, and makes the system easier to explain.

Deterministic components should handle document segmentation, date parsing, section identification, medication lexicons, dose/unit normalisation, ILAE synonym maps, regular expressions for obvious frequencies, negation cues, family-history cues, and JSON/schema validation. They should also manage the output contract: allowed labels, required evidence, valid date formats, valid units, and review flags.

The LLM should be used where the letter requires semantic interpretation: linking a frequency to the correct seizure type, distinguishing current from historical narrative when the wording is subtle, interpreting semiology, handling uncertain or hedged statements, and resolving local contradictions into a structured answer with evidence. It should not be allowed to silently invent labels. It should be asked to cite evidence and express uncertainty.

A suggested high-level pipeline

1. **Deterministic preprocessing**: split sections, anchor dates, identify medication and seizure-term candidates.
2. **Evidence retrieval**: collect sentences and neighbouring context likely to contain phenotype evidence.
3. **Schema-constrained LLM extraction**: output JSON with labels, evidence spans, temporality, as-

sertion, uncertainty, and normalised values.
4. **Deterministic validation**: check JSON validity, allowed ILAE labels, evidence support, date/frequency consistency, medication status, and contradictions.
5. **Review flags**: mark unsupported, contradictory, uncertain, legacy-term, no-evidence, or low-confidence outputs for human review.

This design resonates with general clinical IE work outside epilepsy. Liu et al. (2025) use rule-based filtering and LLM classification for statin information extraction, illustrating how deterministic components can reduce irrelevant text while preserving recall. Dao et al. (2025) describe a schema-oriented generative AI extraction pipeline with validation and guardrails. These papers are useful not because statins or general medical extraction are the same as epilepsy, but because they support the idea that production-grade clinical LLM systems need scaffolding.

8.1 The LLM as a clinical reader, not a database

One risk in LLM-based extraction is that the model behaves like a summariser: it produces what sounds clinically plausible rather than what the text strictly supports. For research extraction, this is dangerous. The output should be disciplined by the letter. If the letter says "possible focal seizures," the output should not say definite focal epilepsy. If the letter says "no convulsions but frequent auras," the output should not collapse that into seizure-free without specifying the event type.

Evidence-span requirements counter this tendency. They force the system to answer: which words in the letter justify the extraction? They also make error analysis more informative. An unsupported answer, a wrong interpretation of correct evidence, and a correct interpretation of wrong evidence are different failures. Without evidence, they all look like the same binary error.

9 Annotation and gold standards: the hidden centre of the dissertation

In a short IEEE paper, annotation often gets compressed into a paragraph. For your dissertation thinking, it should be central. The quality of the system is bounded by the quality and clarity of the target labels. Fonferko-Shadrach, Lacey, Akbari, et al. (2024) make this visible by reporting inter-annotator agreement and describing consensus adjudication. The existence of disagreement is not an embarrassment; it is a measurement of clinical-linguistic difficulty.

For ExECT, you will need to understand the annotation guidelines in detail. Which mentions count? Are negated statements included? How are attributes attached? What happens when a seizure type and an epilepsy type are phrased together? How is certainty encoded? Where is seizure semiology represented, if at all? If your hybrid output schema is richer than the ExECT annotations, you will need a mapping layer for fair evaluation.

For Gan, annotation is embedded differently. The labels are generated or supervised through a task-specific scheme, and the real test set is clinician checked. The relevant question is not just annotation agreement but label grammar: what does each frequency category mean, how are ranges mapped, how is seizure-free duration treated, and how are "unknown" and "no seizure" separated? If you have direct access to the synthetic letters, read the label scheme before reading the model results. The schema is where the clinical assumptions live.

Gold-standard challenge

Before implementing the pipeline, take 20 letters from each benchmark and annotate them manually using the published guidelines. Record every moment when you hesitate. Those hesitations are not a distraction; they are your future error taxonomy, limitations section, and perhaps your best contribution.

10 Evaluation: what reviewers will expect

A strong evaluation should compare deterministic-only, LLM-only, and hybrid systems. Without those baselines, the word ”hybrid” will sound like architecture preference rather than evidence. The ablation should show not only whether the hybrid improves headline F1 but why it improves clinically relevant behaviour.

For the broad ExECT benchmark, report per-entity precision, recall, and F1. Do not hide weak fields behind a global score. Medication and investigation extraction may inflate averages; seizure type, seizure frequency, temporality, and diagnosis certainty are likely more informative for your research question. If your output schema differs from ExECT, specify the mapping and count mapping failures explicitly.

For the Gan seizure-frequency benchmark, report both fine-grained and pragmatic performance if the labels support it. Also report confusion between ”unknown” and ”no seizure,” because that distinction is clinically meaningful. If you normalise to numeric frequency, evaluate both exact or category match and clinically tolerant buckets. A numeric error metric alone may obscure severe semantic failures.

You should also measure properties that classic NLP metrics miss. Unsupported-output rate: how often does the system make a claim with no evidence? Invalid-output rate: how often does the LLM produce malformed JSON or illegal labels? Contradiction rate: how often does the output conflict with another extracted fact? Human-review burden: what proportion of letters are flagged, and are the flags useful?

Table 3: Evaluation dimensions that suit the hybrid framing.

Dimension	Why it matters	Possible metric
Broad phenotype extraction	Shows whether the pipeline handles the full clinic-letter phenotype space rather than one narrow variable.	Per-field precision, recall, F1; macro-F1 across entity types; entity-family scores.
Frequency normalisation	Tests temporal reasoning and clinically useful representation.	Fine-grained label F1, pragmatic label F1, numeric error, bucketed accuracy.
Evidence grounding	Separates correct answers from unsupported plausible answers.	Evidence precision/recall, token-level overlap, unsupported-output rate.
Temporality and assertion	Prevents historical, family-history, negated, and uncertain statements becoming current facts.	Current-vs-historical accuracy; negation accuracy; uncertainty calibration or agreement.
Format reliability	Matters for downstream use and reproducibility.	Valid JSON rate, schema-valid output rate, illegal-label rate.
Clinical severity	Recognises that not all errors are equal.	Error taxonomy with severity categories; clinician review of sampled errors.

10.1 Generalisation and the external-validity question

Generalisation is one of the most important themes in this literature. Decker et al. (2022) show rule-based performance can drop sharply externally. Xie et al. (2023a) examine how fine-tuned transformer models behave in new clinical contexts. Synthetic-data studies must also prove that models trained on synthetic text survive contact with real text (Gan, Roy, et al. 2026; J. J. Li et al. 2021; Melamud and Shivade 2019).

If your dissertation cannot include external real clinical validation, it can still be honest and useful. Use the benchmark pair to test different kinds of generalisation: broad synthetic clinician-authored letters versus task-specific generated synthetic letters; deterministic versus LLM-only versus hybrid; ExECT-style labels versus Gan-style frequency labels. State clearly that this is benchmark generalisation, not deployment validation. That honesty will make the work stronger.

11 Error taxonomy: the part that may make the paper publishable

A good error analysis can make a modest model result interesting. In clinical NLP, reviewers often trust a system more when the authors understand its failures. For seizure phenotype extraction, the error taxonomy should be clinically informed, not just technical.

Table 4: Epilepsy-specific failure modes and how a hybrid system can address them.

Failure mode	Typical example	Hybrid mitigation
Historical vs current seizures	Childhood tonic-clonic seizures extracted as current events.	Section and temporal cue rules; LLM temporality classification; validator requiring current evidence.
No seizure vs unknown	Silence about seizures treated as seizure freedom, or vague text treated as no seizure.	Require explicit evidence for no-seizure labels; allow unknown as valid output.
Multiple event types	Auras weekly, convulsions yearly, absences absent.	Event-type linking; separate frequency objects per seizure type.
Family history	”Mother has epilepsy” extracted as patient diagnosis.	Family-history section/kinship cue rules; assertion-status check.
Functional or non-epileptic attacks	PNES or dissociative seizures mapped to epileptic seizure types.	Controlled labels for non-epileptic events; uncertainty flags; clinician-review category.
Medication status	Stopped or planned medication extracted as current.	Medication-action lexicon; current/past/planned status classifier.
Legacy terminology	”Complex partial” not mapped, or mapped without preserving original text.	ILAE synonym map plus original-surface-text field.
Relative dates	”Six weeks ago” not anchored to letter date.	Date anchor extraction; relative-date normaliser; uncertainty if anchor missing.

Failure mode	Typical example	Hybrid mitigation
Contradictory statements	"No seizures since 2022" and "possible nocturnal event last month."	Contradiction detector; output multiple hypotheses or review flag.

This taxonomy should be populated empirically. During evaluation, sample false positives, false negatives, and near misses. For each, record whether the failure is due to retrieval, interpretation, schema mapping, normalisation, or benchmark ambiguity. This will let you say something more informative than "the model sometimes struggled with complex sentences."

12 The narrative arc for your dissertation paper

The dissertation can be written as a story about complementarity. The opening problem is that epilepsy phenotypes are clinically important but trapped in unstructured letters. The related work shows a progression: rule-based systems such as ExECT prove the value of epilepsy-specific NLP; transformer machine-reading approaches improve seizure outcome extraction; LLMs introduce flexible semantic extraction but raise concerns about unsupported inference; synthetic data offers reproducibility and privacy but needs real-world anchoring. The gap is not "no one has used AI for this." The gap is how to combine deterministic clinical constraints, LLM semantic flexibility, and evidence-grounded evaluation for auditable seizure phenotype extraction.

A concise research claim could be:

Core thesis

This study evaluates whether deterministic clinical NLP components and large language models can be combined to extract auditable, ILAE-aligned seizure phenotypes from epilepsy clinic letters more reliably than either deterministic or LLM-only approaches, using public ExECT annotations for broad phenotype extraction and synthetic seizure-frequency letters for targeted temporal-normalisation benchmarking.

That claim is narrow enough to test and broad enough to matter. It also naturally leads to ablation, error analysis, and clinical discussion.

12.1 A possible argument structure

The paper could be organised around four claims:

1. **Clinical claim**: seizure phenotypes in clinic letters require temporality, assertion, evidence, and schema grounding, not just entity detection.
2. **Methodological claim**: deterministic components and LLMs solve different subproblems and should be combined rather than treated as alternatives.
3. **Benchmark claim**: ExECT and Gan provide complementary reproducible benchmarks for broad and narrow phenotype extraction.
4. **Evaluation claim**: the system should be judged not only by F1 but also by unsupported-output rate, schema validity, temporality errors, and clinical error severity.

This structure would make the literature review serve the methods section rather than sit beside it.

13 Ambiguities to resolve before implementation

Several decisions must be made early, because they affect annotation, prompts, validation, and evaluation.

13.1 Are you extracting explicit documentation or inferred phenotype?

If a letter says "brief staring with lip smacking and unresponsiveness," should the system infer a focal impaired consciousness seizure? A clinician might reasonably infer it, but an annotation benchmark may not label it unless the seizure type is explicit. You need to decide whether the task is extractive, inferential, or a two-stage design: extract semiology first, then optionally map to candidate seizure type with a confidence flag.

13.2 Which ILAE version anchors the schema?

The 2017 ILAE classification remains a common anchor in the literature, but the 2025 update is now relevant. Older letters may contain older terminology, and benchmark annotations may reflect the classification state at the time of design. A versioned mapping would be more defensible than pretending there is one timeless vocabulary.

13.3 What does "current" mean?

Current relative to the letter date? Current relative to the last clinic visit? Current as of the patient's reported history? For medication, current may be clearer than for seizures. For seizure frequency, "current" may cover weeks, months, or since last review depending on context.

13.4 How will contradictions be represented?

Clinic letters can contain contradictory statements because they summarise history, patient report, clinician judgement, and plans. A mature system should not always force one answer. Sometimes the correct output is a review flag with competing evidence spans.

13.5 What is the unit of evaluation?

Mention-level extraction rewards finding all relevant phrases. Letter-level phenotype extraction rewards producing the final structured summary. Patient-level extraction would require longitudinal aggregation. Your dissertation likely sits at the letter-level phenotype summary, but you should state this explicitly.

A deliberately uncomfortable question

If your hybrid system and the benchmark disagree on a clinically ambiguous letter, how will you decide whether this is a model error, a benchmark limitation, or an acceptable alternative reading? The answer should be in your methods, not improvised after seeing results.

14 Reading guide: what to do with the papers next

Read Fonferko-Shadrach, Lacey, Akbari, et al. (2024) slowly with the annotation guidelines open. Do not skim the entity list. Ask why each concept exists, how attributes are attached, where the schema draws boundaries, and which concepts are omitted. Then inspect the Zenodo letters and annotations. Look for cases where the text feels clinically clear but the annotation decision is not obvious. Those are the cases that will teach you the task.

Read Gan, Roy, et al. (2026) with a different mindset. Pay close attention to the label grammar, the synthetic generation design, and the real-test evaluation. Ask how the authors prevent synthetic letters from being merely fluent rather than task-faithful. Look at the evidence spans and rationales as design objects. Then ask what would happen if the same approach were extended beyond seizure frequency to seizure type, medication status, or full seizure phenotype summaries.

Use Xie et al. (2022), Decker et al. (2022), Fernandes, Goldenholz, et al. (2024), Fang et al. (2025), Holgate et al. (2025), and Abeysinghe et al. (2025) as methodological mirrors. Each one solves a related problem with a different balance of annotation, modelling, normalisation, and evaluation. Your task is not to copy one of them. It is to articulate why your hybrid design is the right response to the combined lessons.

15 What would make the final paper feel publishable?

The paper will feel publishable if it does four things well.

First, it must define the extraction target precisely. ”Seizure phenotype” should not float as a vague phrase. Specify which fields are extracted, how they map to ILAE terminology, how temporality and assertion are represented, and what counts as evidence.

Second, it must include meaningful baselines. Deterministic-only and LLM-only systems are not optional; they are necessary to justify the hybrid. Add ablations for evidence retrieval, schema validation, and post-processing if time permits.

Third, it must report per-field and error-type results. Overall F1 is useful but insufficient. The reader should know whether the system succeeds because it extracts medications well or because it truly improves difficult seizure-frequency and temporality fields.

Fourth, it must be honest about generalisation. ExECT and Gan are excellent benchmarks, but they are not proof of safe deployment. A clear limitations section that distinguishes benchmark performance from prospective clinical utility will make the work more credible, not less.

A final design maxim

Make unsupported certainty expensive. In a clinical NLP pipeline, a cautious ”unknown with evidence reviewed” may be more useful than a confident but unsupported label. Your evaluation should reward this discipline.

16 A compact annotated map of the core literature

Benchmark foundations

Fonferko-Shadrach et al. 2024. The public annotated synthetic epilepsy clinic letter corpus. Read this for annotation design, concept boundaries, inter-annotator agreement, and ExECTv2 benchmarking. Its deeper message is that high-quality clinical NLP starts with a carefully negotiated operational definition of the clinical concepts.

Gan et al. 2026. The synthetic seizure-frequency benchmark. Read this for task-faithful synthetic supervision, label grammar, evidence spans, rationales, open-weight model training, and real-letter evaluation. Its deeper message is that synthetic data can be scientifically useful when it is constrained by the task rather than generated as generic clinical prose.

Epilepsy NLP trajectory

Fonferko-Shadrach et al. 2019. The original ExECT system. Read this as the deterministic/rule-based baseline lineage and as evidence that epilepsy clinic letters contain extractable, clinically useful structured data.

Xie et al. 2022. Machine reading for seizure frequency, seizure freedom, and date of last seizure. Read this for task decomposition and the move from entity finding to answer extraction.

Decker et al. 2022. Rule-based seizure type and frequency extraction with poor external generalisation. Read this whenever tempted to claim too much from a single-site evaluation.

Fernandes et al. 2024. RoBERTa plus regex post-processing for seizure-control metrics. Read this as an early hybrid pattern for turning extracted spans into clinically usable categories.

Fang et al. 2025. LLM extraction from epilepsy clinic letters. Read this as the nearest contemporary comparator for LLM-based broad epilepsy information extraction.

Holgate et al. 2025 and Abeysinghe et al. 2025. Recent LLM and pretrained-model work on seizure-frequency extraction. Read these to understand why frequency normalisation remains a live problem even in the LLM era.

Clinical vocabulary and reporting scaffolds

ILAE 2017/2022/2025 classification and terminology papers. Read these to avoid collapsing seizure type, epilepsy type, syndrome, semiology, and aetiology into one undifferentiated label space.

TRIPOD-LLM and clinical LLM extraction papers. Read these to strengthen reporting transparency, prompt documentation, data provenance, validation, and guardrail descriptions (Collins et al. 2025; Dao et al. 2025; Liu et al. 2025).

17 Closing synthesis

The literature points toward a simple but demanding conclusion: epilepsy phenotype extraction is not just a natural language problem. It is a clinical representation problem. The system must read text, but it must also respect the structure of epilepsy knowledge, the uncertainty of clinical documentation, the temporality of seizure histories, and the privacy constraints of health records.

The ExECT annotation corpus gives you the broad public benchmark needed for reproducibility. The Gan synthetic-letter work gives you a focused and ambitious benchmark for seizure-frequency reasoning under privacy constraints. Together, they allow a dissertation to ask a nuanced question: can a hybrid deterministic plus LLM pipeline preserve the auditability and schema discipline of traditional clinical NLP while gaining the interpretive flexibility of modern language models?

That question is worth pursuing because it is not a fashion statement about LLMs. It is a practical question about how clinical AI systems should be built when the information is important, messy, private, and only partly reducible to rules.

Full reference links

Abeysinghe, R., C. Tao, J. Li, et al. (2025). "Leveraging pretrained language models for seizure frequency extraction from epilepsy evaluation reports". In: *npj Digital Medicine* 8, p. 439. DOI: 10.1038/s41746-025-01592-4. URL: https://doi.org/10.1038/s41746-025-01592-4.

Beniczky, S., E. Wirrell, et al. (2025). "Updated classification of epileptic seizures: Position paper of the International League Against Epilepsy". In: *Epilepsia*. DOI: 10.1111/epi.18338. URL: https://doi.org/10.1111/epi.18338.

Blumcke, I. et al. (2022). "Updated and expanded terminology for organization of seizures and epilepsies: report of the ILAE Task Force on Classification and Terminology". In: *Epileptic Disorders* 24.3, pp. 447–456. DOI: 10.1684/epd.2022.1430. URL: https://doi.org/10.1684/epd.2022.1430.

Collins, G. S., P. Dhiman, C. L. Andaur Navarro, et al. (2025). "TRIPOD-LLM: a reporting guideline for studies using large language models". In: *Nature Medicine* 31, pp. 60–69. DOI: 10.1038/s41591-024-03425-5. URL: https://doi.org/10.1038/s41591-024-03425-5.

Dao, N., L. Quesada, S. M. Hassan, et al. (2025). "Generative artificial intelligence for automated data extraction from unstructured medical text". In: *JAMIA Open* 8.5, ooaf097. DOI: 10.1093/jamiaopen/ooaf097. URL: https://doi.org/10.1093/jamiaopen/ooaf097.

Decker, B. M., C. E. Hill, A. K. Shah, et al. (2022). "Development of a natural language processing algorithm to extract seizure types and frequencies from the electronic health record". In: *Seizure* 101, pp. 43–49. DOI: 10.1016/j.seizure.2022.07.010. URL: https://doi.org/10.1016/j.seizure.2022.07.010.

Fang, B., A. Akbari, W. O. Pickrell, et al. (2025). "Extracting epilepsy-related information from unstructured clinic letters using large language models". In: *Epilepsia*. DOI: 10.1111/epi.18475. URL: https://doi.org/10.1111/epi.18475.

Fernandes, K., D. M. Goldenholz, et al. (2024). "Extracting seizure control metrics from clinic notes of patients with epilepsy: a natural language processing approach". In: *Epilepsy Research* 204, p. 107451. DOI: 10.1016/j.eplepsyres.2024.107451. URL: https://doi.org/10.1016/j.eplepsyres.2024.107451.

Fisher, R. S. et al. (2017). "Operational classification of seizure types by the International League Against Epilepsy: Position Paper of the ILAE Commission for Classification and Terminology". In: *Epilepsia* 58.4, pp. 522–530. DOI: 10.1111/epi.13670. URL: https://doi.org/10.1111/epi.13670.

Fonferko-Shadrach, B., A. S. Lacey, A. Akbari, R. A. Lyons, M. I. Rees, and W. O. Pickrell (2024). "Annotation of epilepsy clinic letters for natural language processing". In: *Journal of Biomedical Semantics* 15.1, p. 22. DOI: 10.1186/s13326-024-00316-z. URL: https://doi.org/10.1186/s13326-024-00316-z.

Fonferko-Shadrach, B., A. S. Lacey, and W. O. Pickrell (2023). *Synthetic epilepsy clinic letters and annotations for natural language processing*. Zenodo. DOI: 10.5281/zenodo.8381080. URL: https://zenodo.org/records/8381080.

Fonferko-Shadrach, B., A. S. Lacey, C. P. White, et al. (2019). "Using natural language processing to extract structured epilepsy data from unstructured clinic letters: development and validation of the ExECT system". In: *BMJ Open* 9.4, e023232. DOI: 10.1136/bmjopen-2018-023232. URL: https://doi.org/10.1136/bmjopen-2018-023232.

Gan, J., S. Roy, et al. (2026). *Reproducible Synthetic Clinical Letters for Seizure Frequency Information Extraction*. Preprint. arXiv: 2603.11407. URL: https://arxiv.org/abs/2603.11407.

Holgate, J. et al. (2025). "Fine-tuning LLMs to Extract Epilepsy Seizure Frequency Data from Health Records". In: *Proceedings of the BioNLP Workshop*. Association for Computational Linguistics. DOI: 10.18653/v1/2025.bionlp-1.5. URL: https://aclanthology.org/2025.bionlp-1.5/.

Li, J. J., S. Fu, J. Chen, et al. (2021). "Assessing the utility of synthetic clinical notes for clinical natural language processing: a pilot study". In: *Journal of the American Medical Informatics Association* 28.11, pp. 2473–2479. DOI: 10.1093/jamia/ocab112. URL: https://doi.org/10.1093/jamia/ocab112.

Liu, S., A. B. McCoy, Q. Chen, et al. (2025). "Integrating rule-based NLP and large language models for statin information extraction from clinical notes". In: *International Journal of Medical Informatics* 204, p. 106104. DOI: 10.1016/j.ijmedinf.2025.106104. URL: https://doi.org/10.1016/j.ijmedinf.2025.106104.

Melamud, O. and C. Shivade (2019). "Towards automatic generation of shareable synthetic clinical notes using neural language models". In: *Proceedings of the 2nd Clinical Natural Language Processing Workshop*, pp. 35–45. DOI: 10.18653/v1/W19-1905. URL: https://doi.org/10.18653/v1/W19-1905.

Ojemann, J. G., F. Xie, et al. (2025). "Zero-Shot Extraction of Seizure Outcomes from Clinical Notes Using Generative Pretrained Transformers". In: *Journal of Healthcare Informatics Research*. DOI: 10.1007/s41666-025-00198-5. URL: https://doi.org/10.1007/s41666-025-00198-5.

Scheffer, I. E. et al. (2017). "ILAE classification of the epilepsies: Position paper of the ILAE Commission for Classification and Terminology". In: *Epilepsia* 58.4, pp. 512–521. DOI: 10.1111/epi.13709. URL: https://doi.org/10.1111/epi.13709.

Xie, F. et al. (2023a). "Generalization of finetuned transformer language models to new clinical contexts". In: *JAMIA Open* 6.3, ooad070. DOI: 10.1093/jamiaopen/ooad070. URL: https://doi.org/10.1093/jamiaopen/ooad070.

— (2023b). "Long-term epilepsy outcome dynamics revealed by natural language processing of clinic notes". In: *Epilepsia* 64.10, e173–e179. DOI: 10.1111/epi.17633. URL: https://doi.org/10.1111/epi.17633.

Xie, F., L.-W. H. Lehman, V. Manolaki, et al. (2022). "Extracting seizure frequency from epilepsy clinic notes: a machine reading approach to natural language processing". In: *Journal of the American Medical Informatics Association* 29.5, pp. 873–881. DOI: 10.1093/jamia/ocac018. URL: https://doi.org/10.1093/jamia/ocac018.

Yew, R., S. Chae, R. Arablouei, et al. (2023). "Transforming epilepsy research: a systematic review on natural language processing applications". In: *Epilepsia* 64.2, pp. 292–305. DOI: 10.1111/epi.17474. URL: https://doi.org/10.1111/epi.17474.